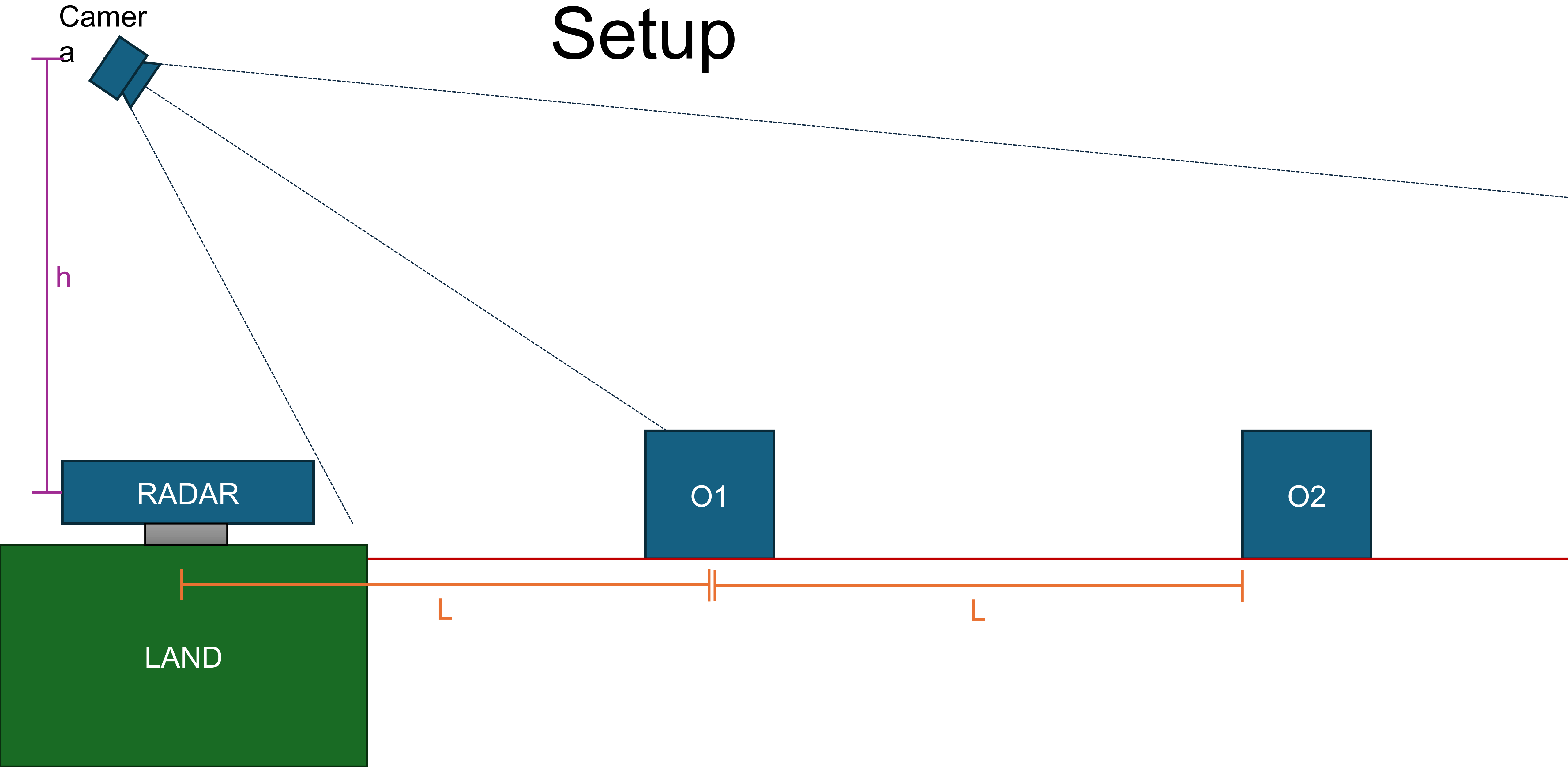
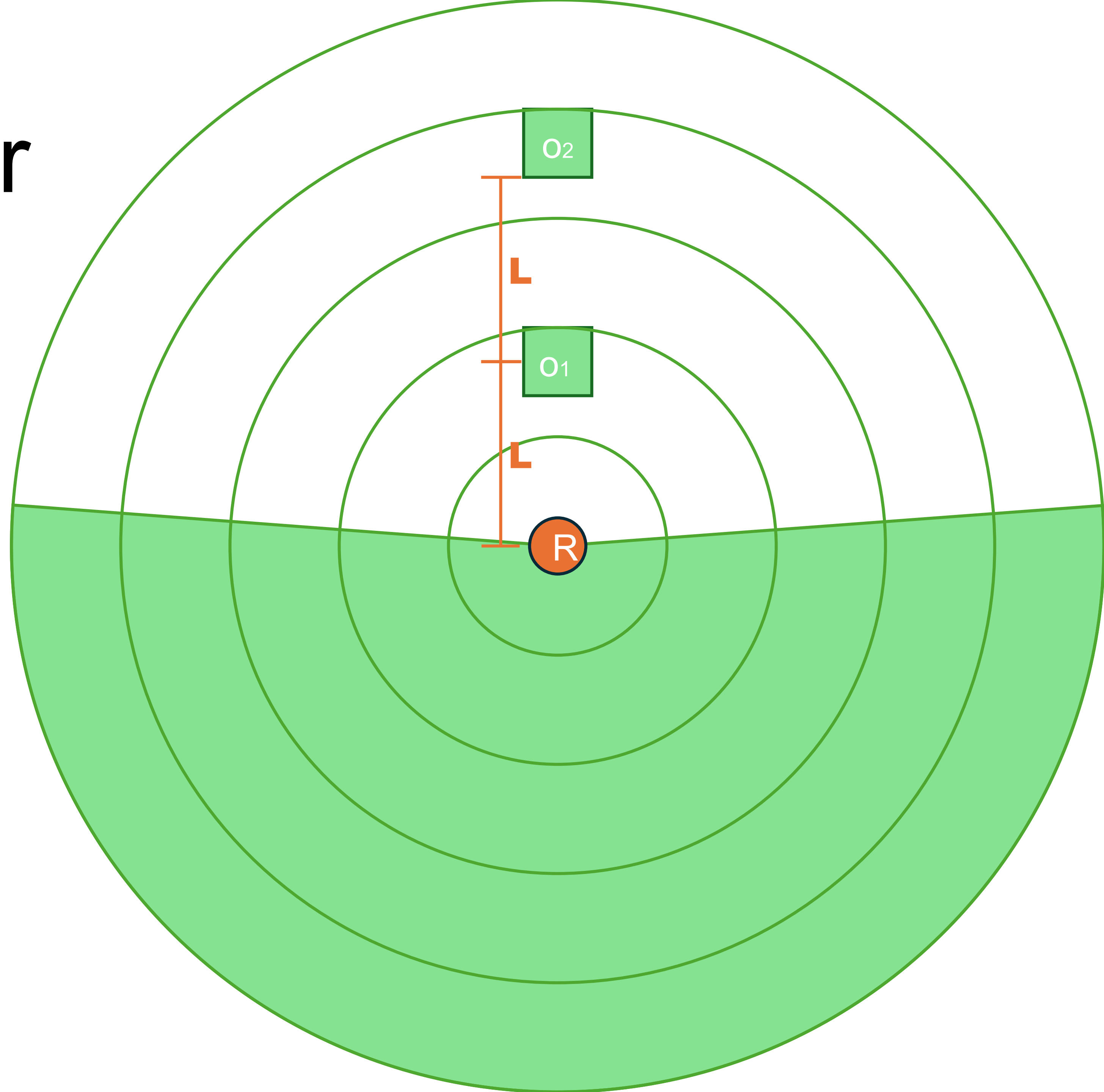
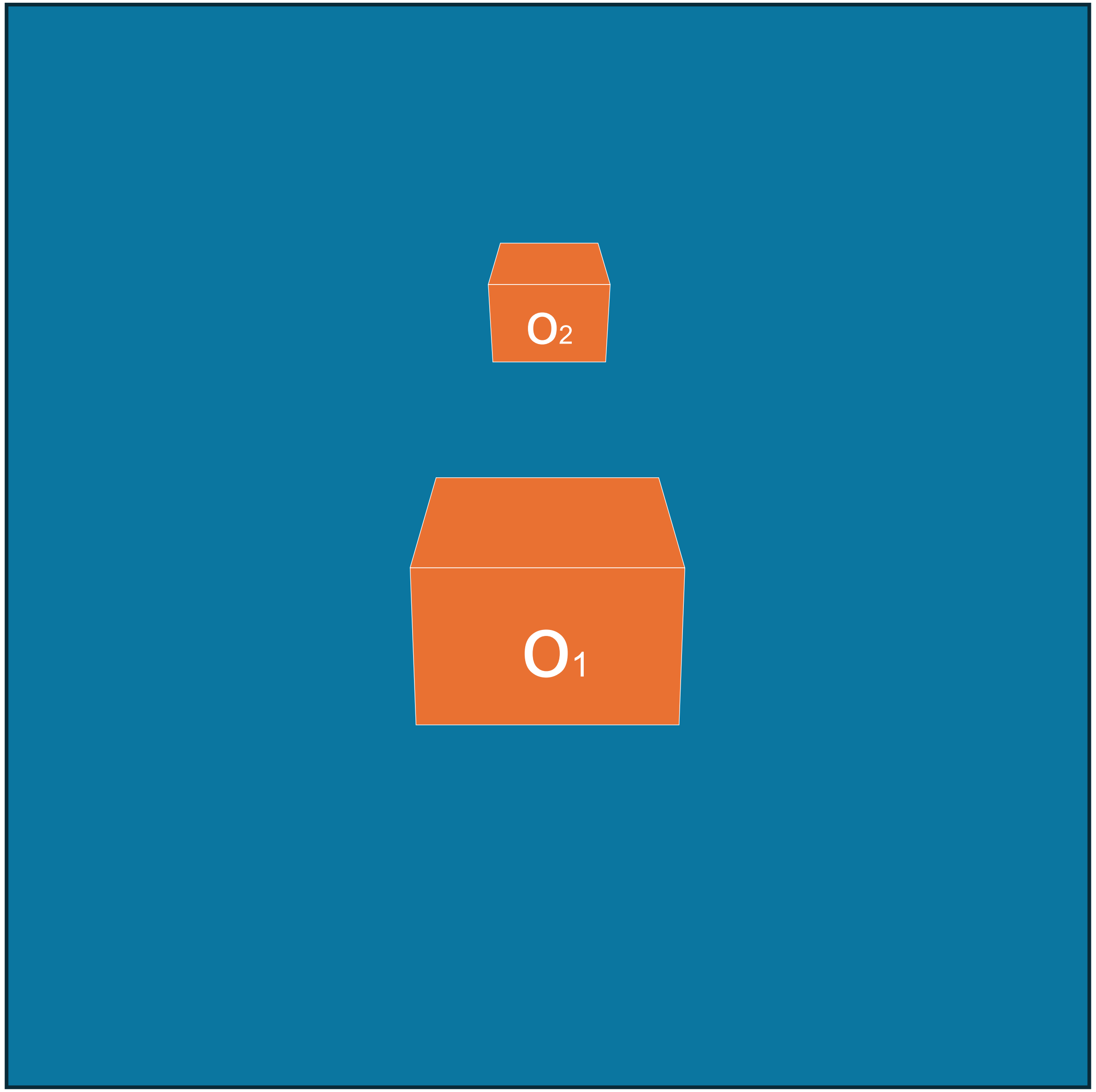
SYSTEM Setup



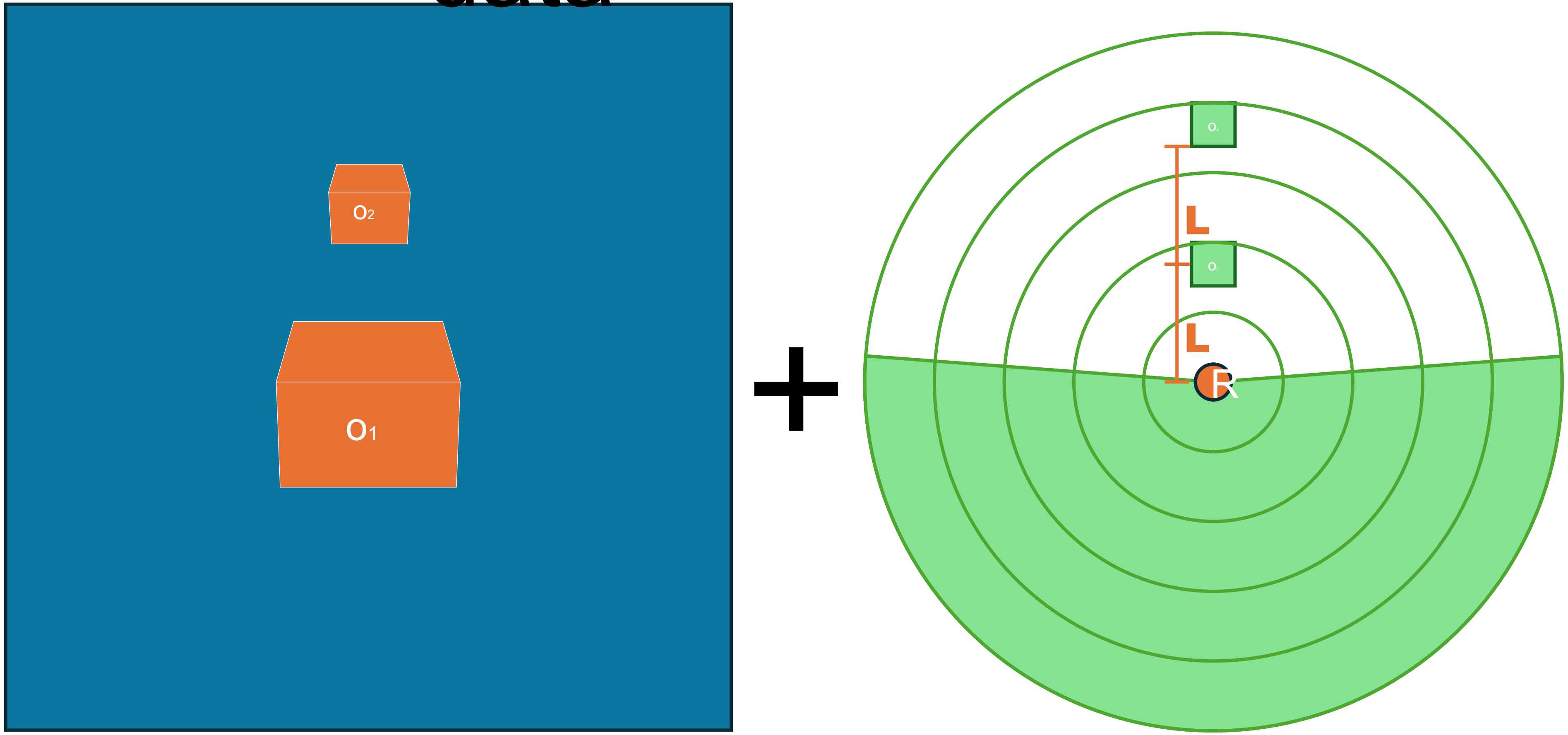
Output from Radar (ideal)



Output from Camera (ideal)



Combine the data

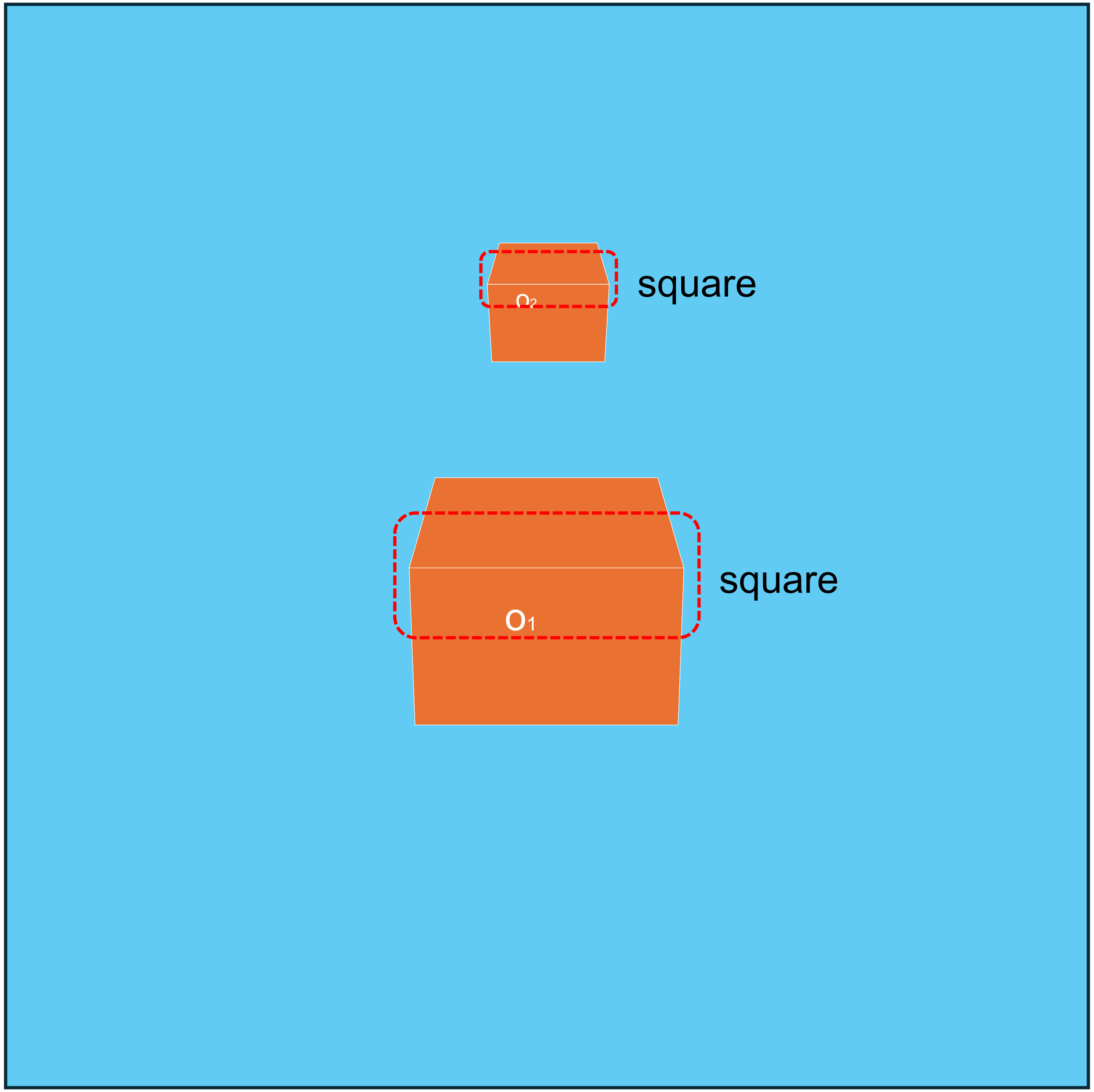


Radar-overlay ONLY

Radar only has 2-dimensional plane to look at, and does not have a sense for 3d (i.e. height)

The bounding boxes are overlayed to account for camera distortion of far away objects

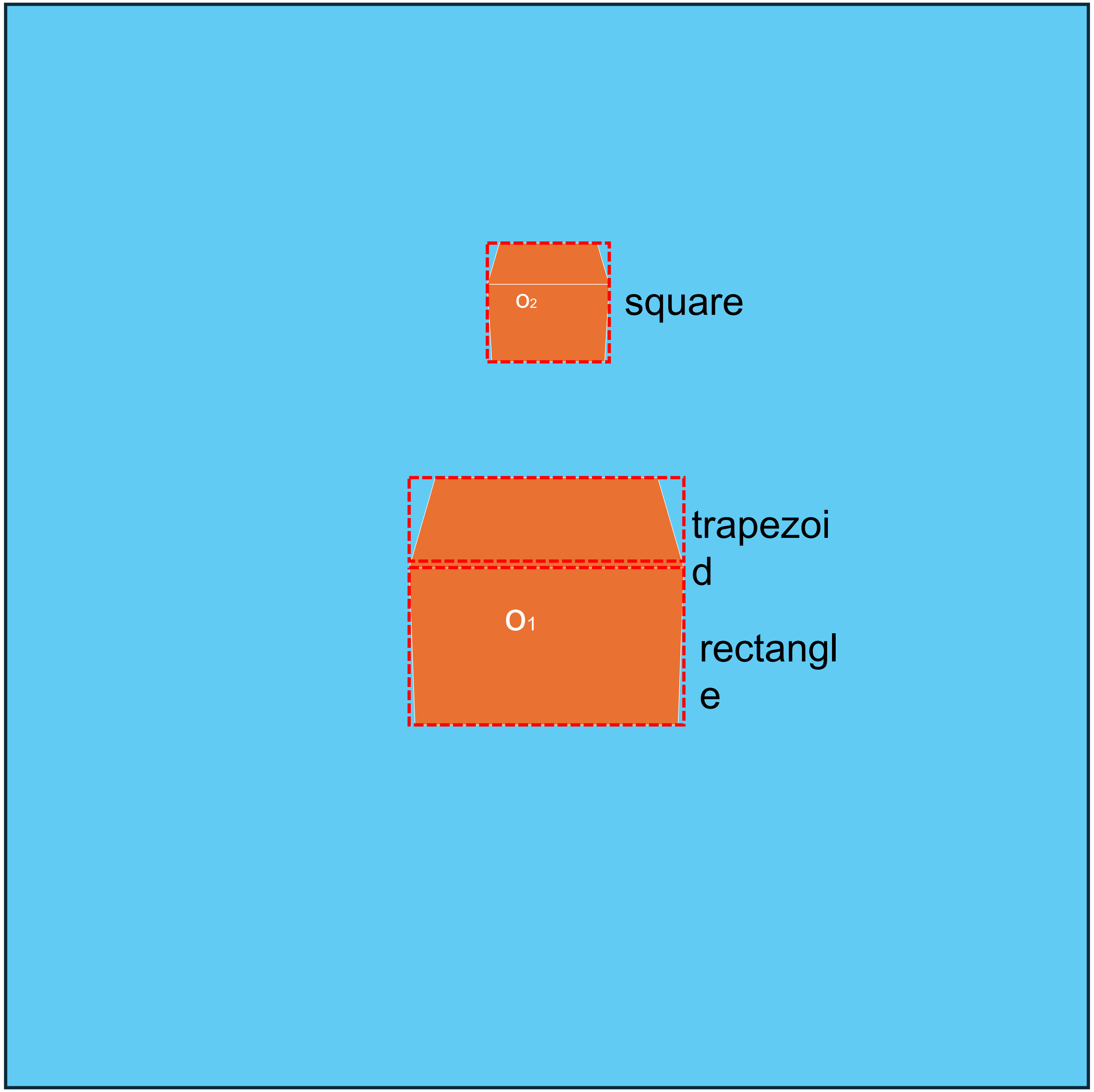
Can only identify as ‘square’ in this case because of the 3d limitations



AI camera-overlay only

Only has the camera view. Can not see a cross-sectional area of object.

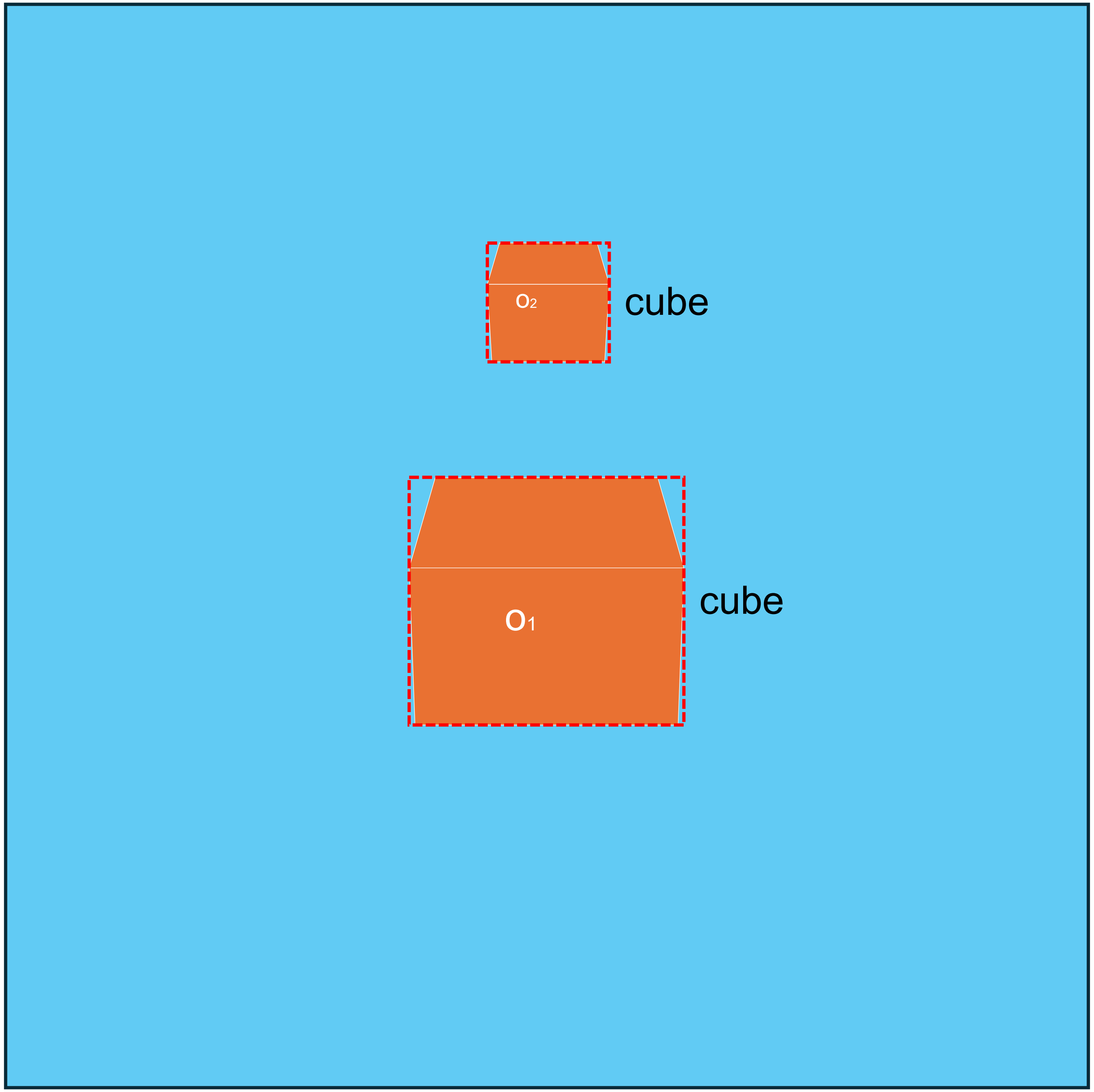
Can lead to some incorrect assumptions, i.e. Trapezoid and rectangle



AI + Radar Overlay

By combining AI and radar, object identification becomes more accurate and the labeling process is accelerated.

Additionally, this integration enables the detection of objects that appear only a few pixels wide in the camera system but are detectable by radar.



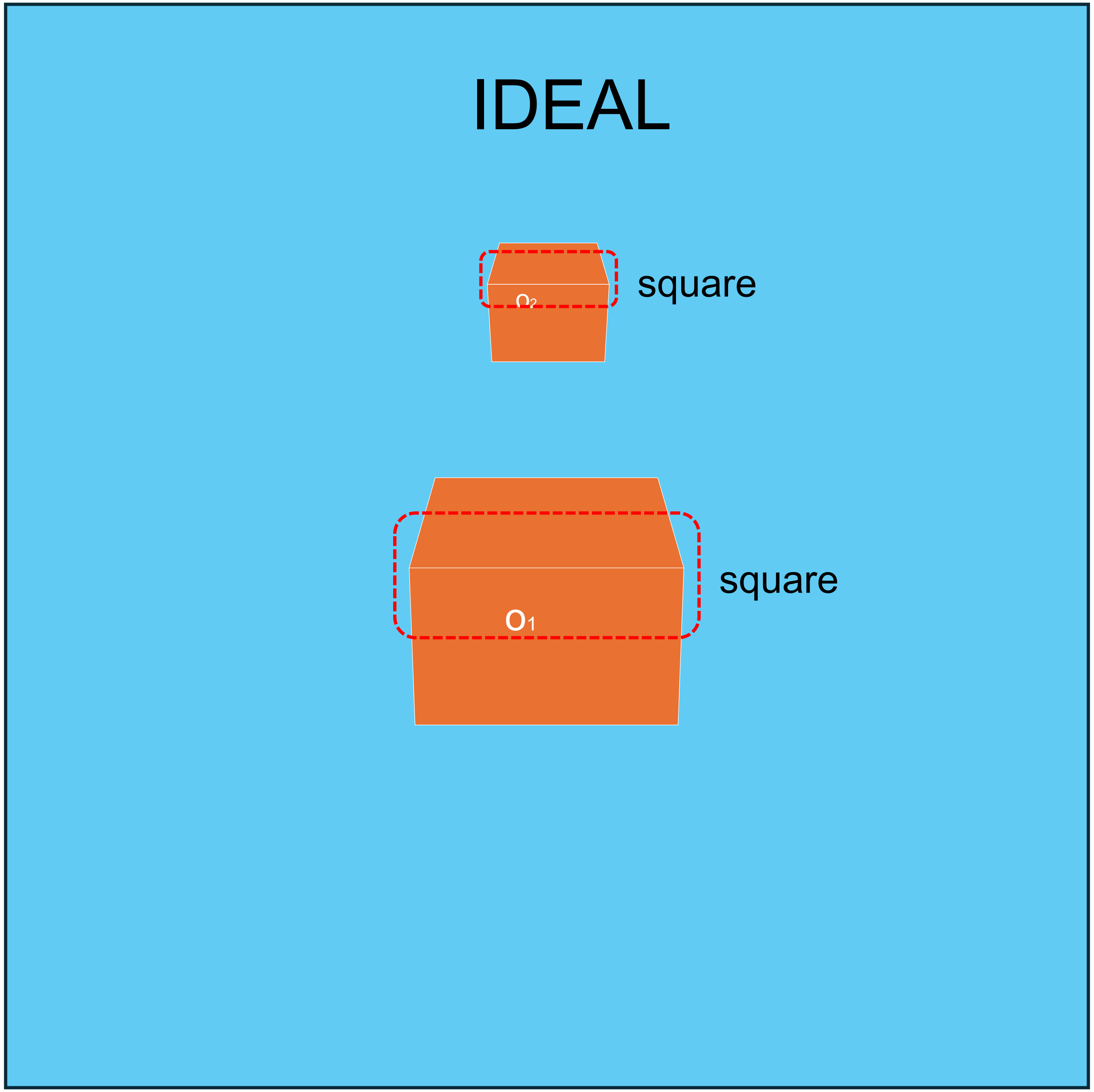
My Goals

Radar Overlay and solutions

Radar-overlay ONLY

My goal is to automate the Radar process so that the task of overlaying can be easier.

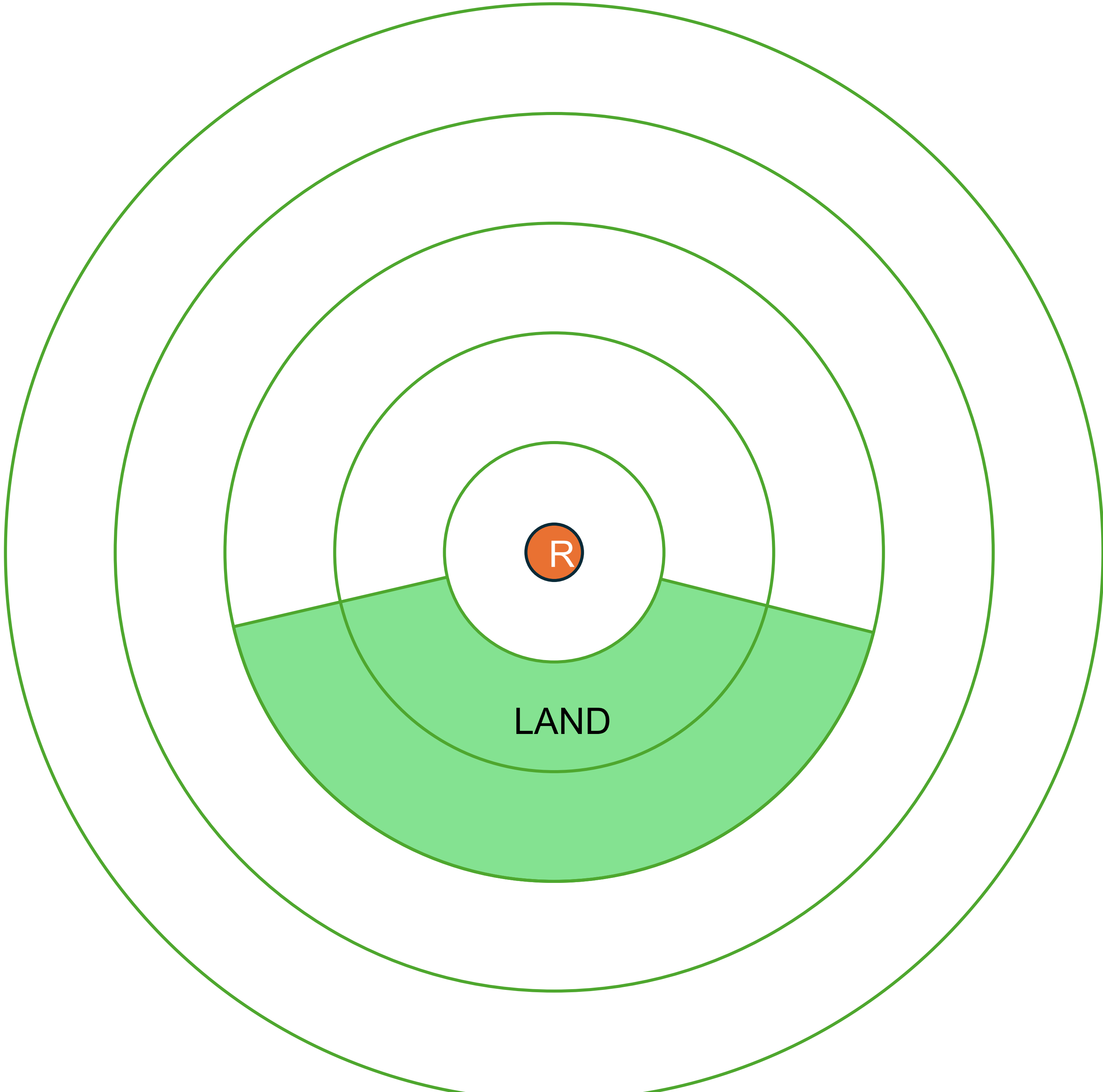
In radar as produced by manufacturer has a very steep cut-off range.



Radar-overlay ONLY

For example, if the gain is set too low, then the radar would show almost no detected object.

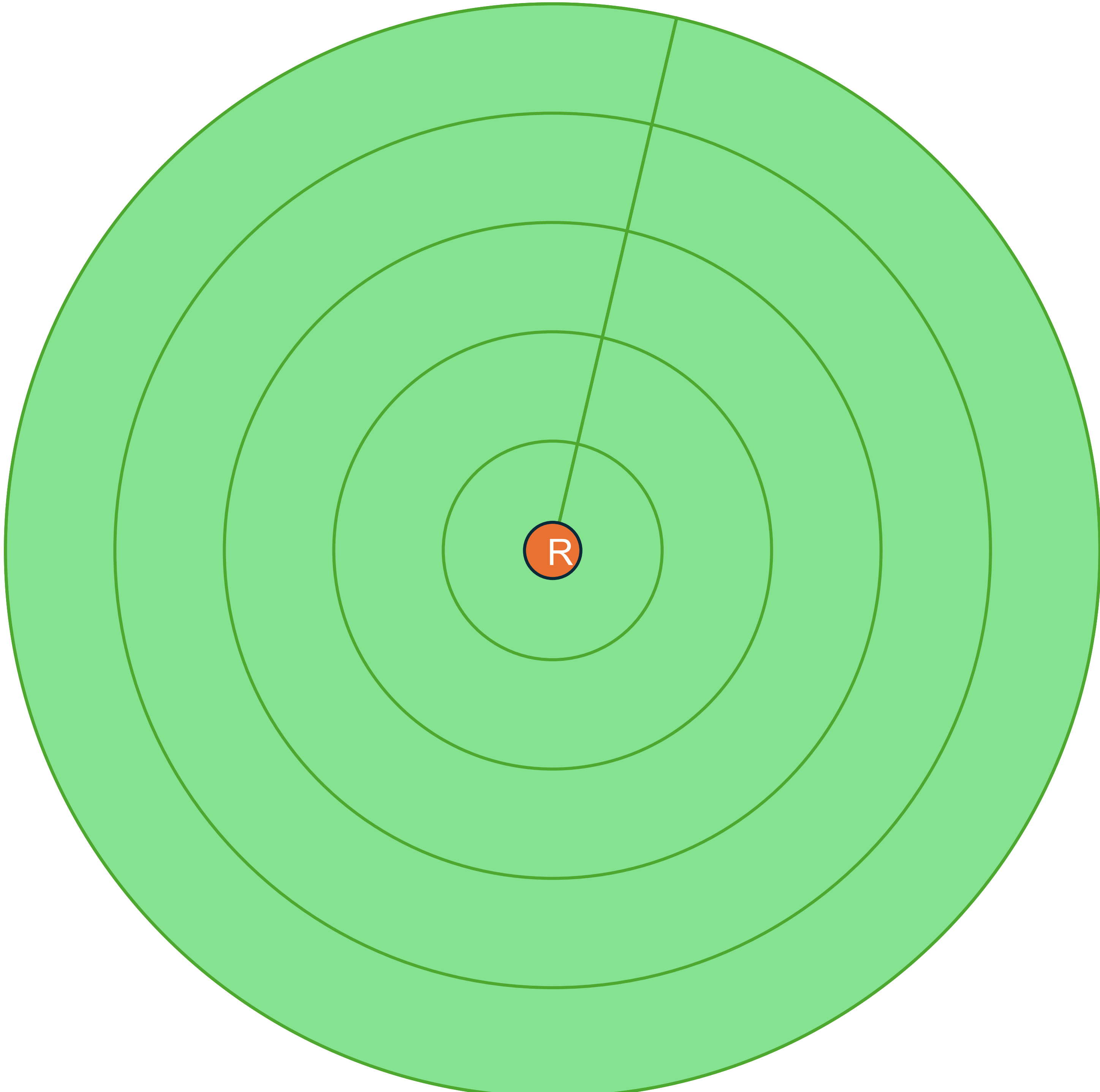
Note how even the land doesn't get detected in some areas.



Radar-overlay ONLY

Additionally, if the gain is set too high, then the radar will show that everything is being detected(including noise and other interference).

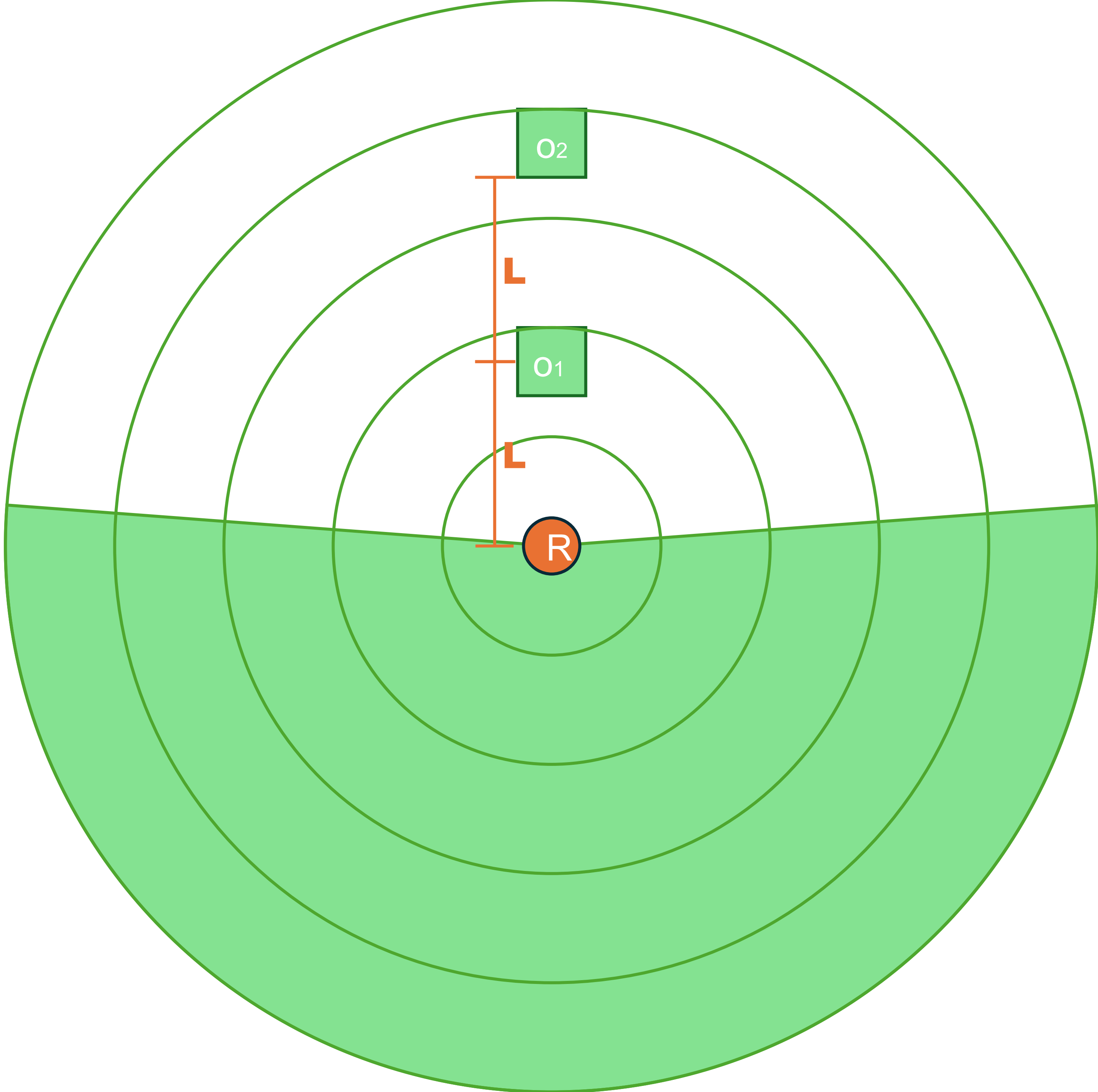
Note how now the objects can’t be detected at all because everything else is being detected.



Radar-overlay ONLY

Therefore, in most scenarios, the operator must set the gain manually to get the best results.

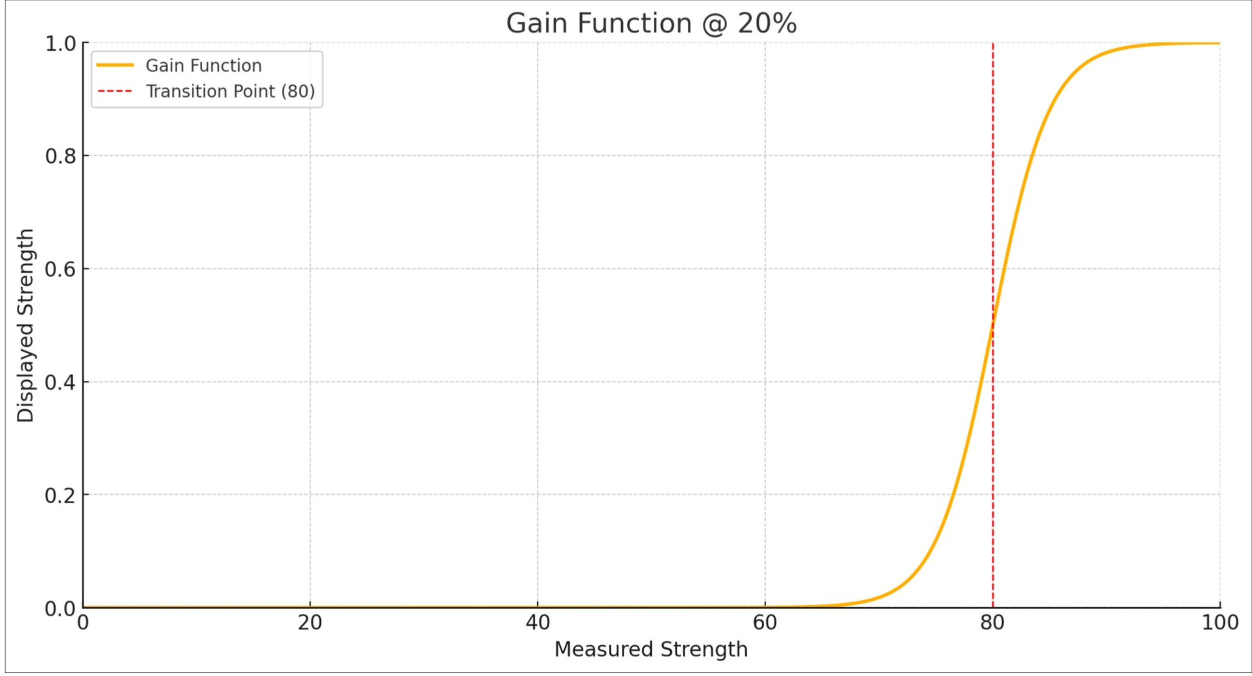
However, to make the data labelling process easier and to reduce reliance on trained operators, an automatic system is needed.

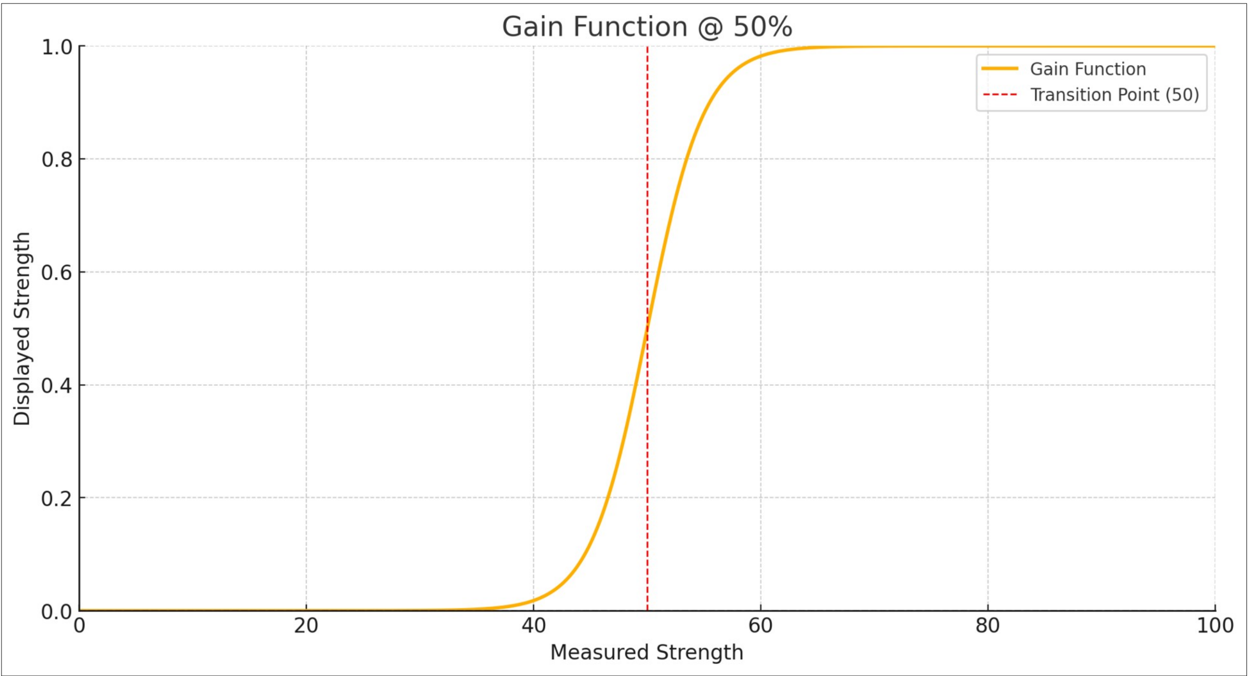


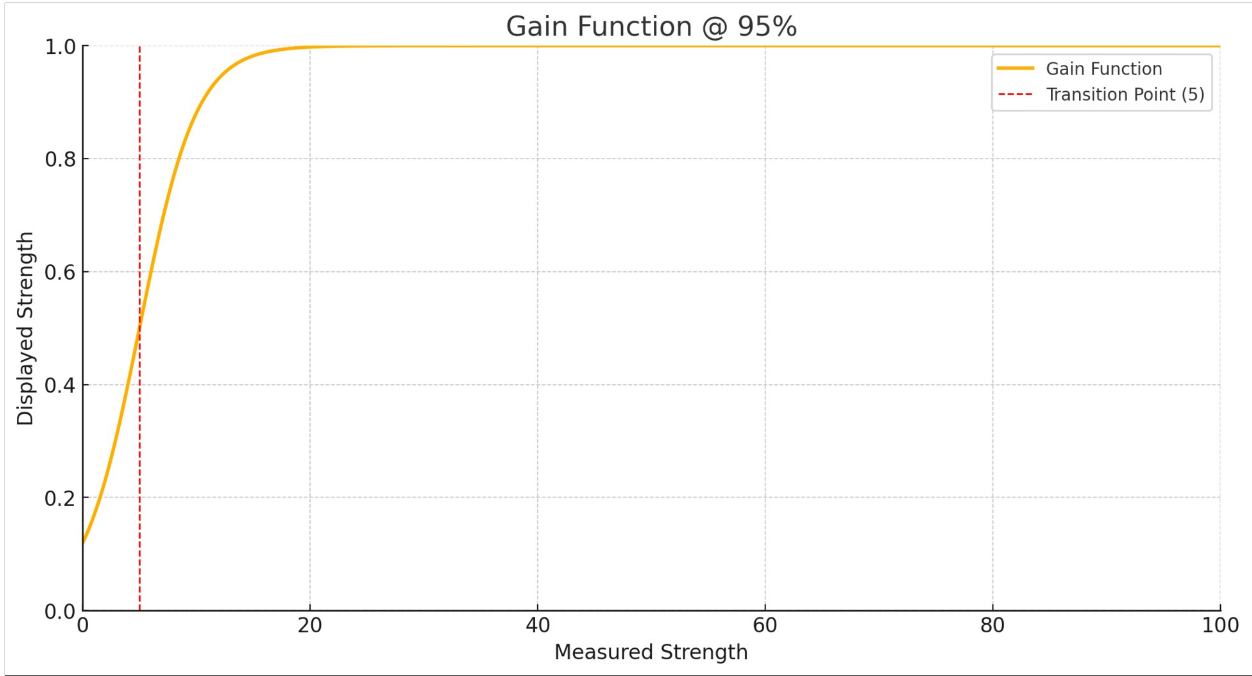
Radar-overlay ONLY

- This is a mockup of how the radar functions seem to work (real vs displayed strength)

- Note how most values(normalized) will either be a 1 or 0 regardless of the gain setting due to the activation function of the radar.

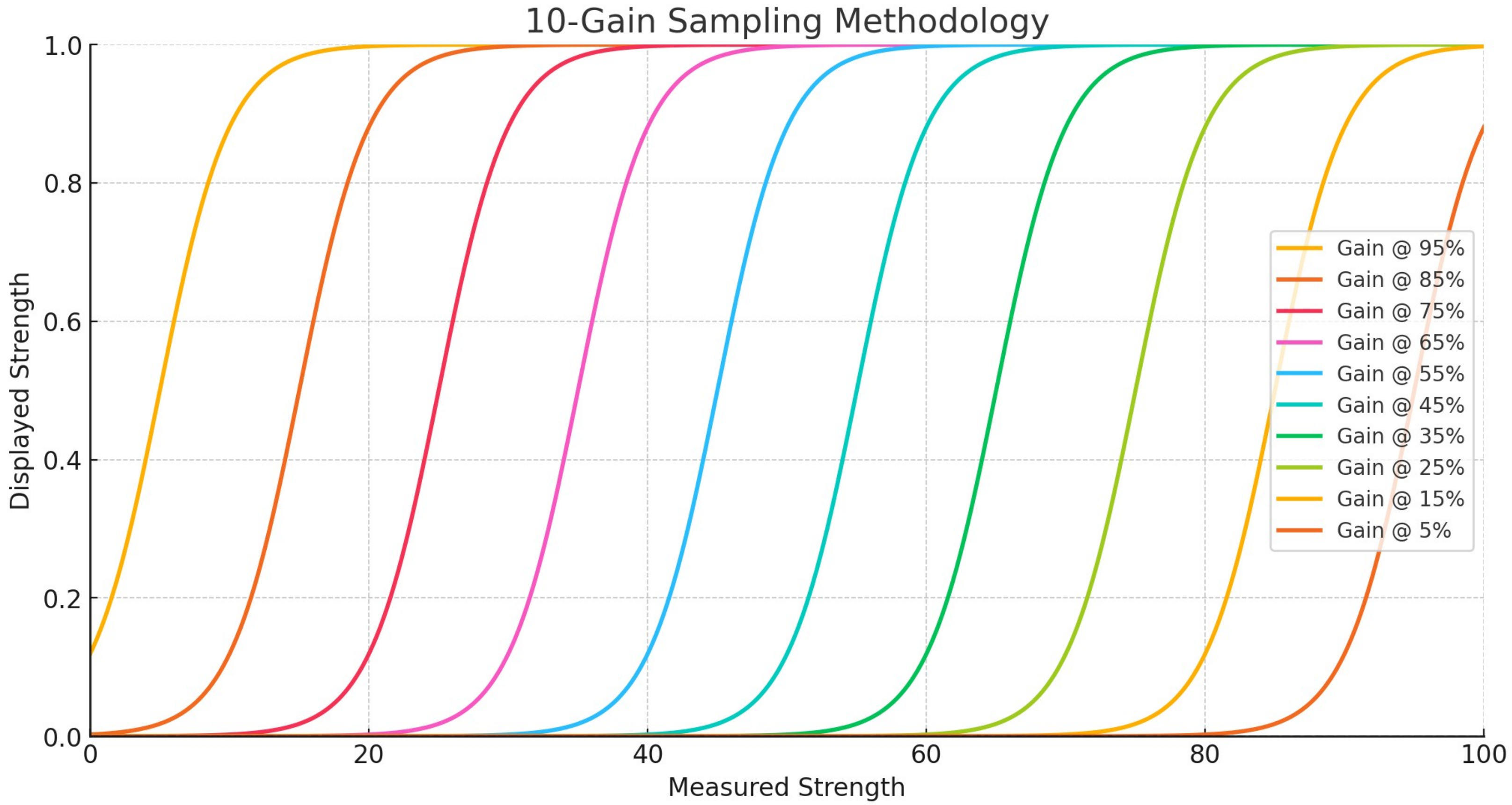






Radar-overlay ONLY

Since the system tends to prioritize 0 or 1s, then by combining multiple ‘frames’ at different gains, one can gain a better approximation of the true signal strength.



Radar-overlay ONLY

For object detection purposes, feeding multiple ‘frames’ @ different gains into a CNN would likely be ideal to detect the objects from the radar.

Final result should be a radar frame with labelled objects.